



Full-Featured Interactive Web Application Specification



Project Overview

EcoByte is an interactive web application designed to help users understand and reduce their digital carbon footprint. Through an engaging chatbot experience, users discover the environmental impact of their digital habits and receive personalized recommendations to reduce their ecological impact.



1. Theme and Style

Color Palette

Soft pastel cloud and landscape backgrounds with green, blue, and white tones

Typography

Clean, friendly, modern sans-serif for all headings and text

UI Elements

Rounded, friendly, modern interface with shadowed cards

Branding

EcoByte logo prominently displayed using generated vector logo

- **Background:** Soft hills and clouds landscape
- **Cards:** Rounded corners with soft shadows for content areas
- **Buttons:** Rounded design with soft shadows and hover animations
- **Overall Feel:** Welcoming, modern, and eco-friendly aesthetic



2. Welcome Screen

Welcome Message Content

"Welcome to EcoByte Understand Your Digital Footprint Our digital habits impact the planet. Here, you can make the invisible visible by showing the energy, carbon, and water used by your online activities. The first step toward impact is the will to act. Answer a few simple questions and our assistant will help you see – and reduce – your digital carbon footprint."
[Button: Start Chatting]

Layout Features:

- **Full-screen interface** with centered introduction text
- **Chatbot avatar** at bottom right corner (small avatar with leaf, friendly AI look)
- **Background:** Landscape with soft hills and clouds
- **Interactive chatbot** with input field
- **Initial chatbot message:** "Ready to begin the journey? I'll guide you through a few questions to reveal your digital impact."
- **Behavior:** Welcome text disappears after first interaction



3. User Inquiry / Chatbot Interaction

Question Flow:

- **Step 1:** Chatbot asks for user's name (for engagement)
- **Step 2:** Chatbot asks for user's country
- **Step 3:** Guided questions about digital habits

Digital Habits Questions (Examples):

- Average screen time per day (hours)
- Emails received/sent per day
- Cloud storage usage (GB/photos/videos)
- Streaming hours & quality
- Devices used daily

UX Features:

- **Open-ended answers:** Users can type their responses freely
- **No invasive permissions:** No account connections required
- **Progress bar:** Shows completion status at top of central card
- **Question history:** Display each previous question with the user's trimmed/filtered answer
- **Current question display:** Only show active question in chatbot bubble

Example Chatbot Tone:

"On average, how many emails do you receive per day? (Work, university, newsletters, spam — everything.) You can answer approximately, for example: 20, 50, 100..."

Tone: Friendly, encouraging, non-judgmental



4. Estimation & Visualization

AI Calculations (After Submission):

- **Carbon footprint:** kg CO₂
- **Energy consumption:** kWh
- **Water usage:** liters

Real-World Equivalents:



Homes Powered

Convert energy to "homes powered for X days"



Water Usage

Equivalent to Y Olympic pools or bottles



Electricity

Power for iconic building in user's country
(e.g., Eiffel Tower for 17 hours)

Display Requirements:

- **Interactive cards** with icons and simple numbers
- **Avoid raw numbers alone** - always contextualize
- **Real-world estimations** must be calculated precisely
- **Update animation:** When users complete to-do actions, recalculate and animate UI updates
- **Default location:** Use Eiffel Tower if AI can't determine user's country landmark



5. Interactive To-Do List

Layout:

- **Position:** Left-hand card on dashboard
- **Content:** Personalized actions based on user's previous inputs
- **Status tracking:** Checkbox for "done / not done"

Action Card Example:

Action: "Delete duplicate photos & videos"

Estimated Impact: ~3 GB cloud, 0.04 kg CO₂/month

Status: ☐ Not Done

Popup Modal (When Action Clicked):

- **Purpose:** Why this action matters
- **Impact / What You Save:** Storage, CO₂, energy, water breakdown
- **Step-by-step Guide:** How to complete the action
- **Close Button:** Return to main dashboard

Behavior:

- When marked as done → **recalculate totals**
- Update visual cards with **animation**
- Reflect reduced footprint in real-time



6. Behavior and Workflow

- **State Management:** Maintain user inputs in memory for calculations
- **Dynamic Recalculation:** Update footprint when actions marked as done
- **Smooth Transitions:** Subtle animations for all cards and popups
- **Chatbot Accessibility:** Always visible at bottom right, encourages engagement
- **Privacy First:** No personal accounts or sensitive info required



7. Responsiveness

- **Primary Target:** Desktop-first design
- **Scalability:** Should scale nicely for tablets and large screens
- **Layout:** Clean layout even with large user data
- **Avoid Clutter:** Maintain visual harmony at all screen sizes



8. Assets

- Use all provided screenshots and generated images for reference
- Include EcoByte logo from provided generation
- Landscape/cloud background theme for continuity across screens
- Maintain consistent visual identity throughout the application



9. Workflow / User Flow

Step 1: Welcome Screen

Show intro → chatbot asks to begin

Step 2: Chatbot Questions

Input answers → show progress + previous answers

Step 3: Submit & Calculate

AI calculates footprint → display real-world equivalents on cards

Step 4: To-Do List Interaction

Display To-Do list → user clicks action → open popup → mark as done → update totals

Step 5: Continuous Engagement

All changes dynamically update footprint visuals

Step 6: Ongoing Support

Chatbot always accessible for guidance or reminders



10. Additional Notes

Core Principles:

- **Playful & Educational:** Keep interactions motivating, not guilt-inducing
- **Minimalistic yet Powerful:** Numbers are contextualized, animations are subtle
- **Privacy-Focused:** No personal accounts or sensitive info needed
- **User-Centric Design:** Prioritize clarity, engagement, and visual harmony
- **Accessibility:** Ensure all users can understand and interact with content
- **Transparency:** Show clear calculations and real-world impacts

UX Decision Framework:

- All UX decisions should prioritize clarity
- Maintain engagement through friendly tone and visual feedback
- Visual harmony with existing EcoByte screenshots
- Educational without being preachy
- Empower users to take action



Success Criteria

- ✓ Users complete the chatbot questionnaire with ease
- ✓ Environmental impact is clearly visualized and understood
- ✓ Users feel motivated (not guilty) to reduce digital footprint
- ✓ To-do actions are personalized and actionable
- ✓ UI is visually cohesive with EcoByte brand identity
- ✓ All calculations are scientifically accurate
- ✓ User privacy is fully protected
- ✓ Experience is smooth, engaging, and educational